

FYP Pre-Start Prep — Progress Tracker

LLM-Based Interface for Robotics Systems using UR5e | Week 4 onwards | simulation on MuJoCo
Status snapshot as of Sunday 12 July 2026 (mid Week 4) · Supervisor: Prof Cheah · Prep period 16 Jun – 4 Aug 2026

Legend: [x] done [~] in progress [] to do

Where you stand

The foundation (Weeks 1–3) and the Architecture-B *data-capture* half are done and slightly ahead of schedule. The only item still open this week is the Open X-Embodiment reading. Not yet started: all of Architecture-B training (Week 5) and all of the Architecture-A modular pipeline (Week 6) — the two halves the whole comparison rests on. One code-level gap to close early: `src/fyp/transforms.py` is empty.

WEEK 4 · 7–13 Jul 2026 — Deep VLA reading + MuJoCo simulation

Done 21 · In progress 1 · To do 2 · 24 items

Plan note — *simulation moved from ManiSkill3 to MuJoCo. The sim stack was built directly in MuJoCo (`mujoco_controller`, `ik`, `sim_server/client`, `gripper attach`) — lighter, self-owned, and it provides the RGB-D camera + physics that URSim lacks, which Week-6 perception depends on.*

Code gap — *`src/fyp/transforms.py` is empty. The pose helpers (`pose_to_T`, `T_to_pose`, `pose_trans`) were in an old `square.py`, deleted from git but recoverable (commit 99541bf); the quat-to-rotvec conversion is duplicated across `ik.py`, `mujoco_controller` and `sim_server`. Recover the first and centralise the second before Week-6 hand-eye calibration.*

Mon 7 Jul — RT-2 review + transition to OpenVLA

- [x] Re-read RT-2 notes: what the action space is exactly
- [x] Map RT-2's action space onto what your URController accepts
- [x] Read OpenVLA introduction and related work
RT-1 and RT-2 already deep-read with architectural walkthroughs in Week 3.

Tue 8 Jul — OpenVLA architecture

- [x] OpenVLA Sec 3–4: Llama-2 backbone, DINOv2 + SigLIP vision encoders
- [x] Note action representation and chunking; compare architecturally to RT-2

Wed 9 Jul — OpenVLA training and eval

- [x] OpenVLA Sec 5–6: Open X-Embodiment training subset, LoRA fine-tune recipe, limitations
The LoRA recipe is the one you'll mirror with SmolVLA.

Thu 10 Jul — Open X-Embodiment

- [] Read the OXE paper / data card (Padalkar et al. 2023); action/observation formats; check UR5 inclusion
The one Week-4 reading item still outstanding.

Fri 11 Jul — MuJoCo simulation setup (was ManiSkill3)

- [x] Install MuJoCo; load the UR5e model (Menagerie `ur5e.xml` + `scene.xml`)
- [x] Launch the viewer; verify rendering (X11 for viewer, EGL for headless)
- [x] Attach the Robotiq 2F-85 via `mjSpec` to `scene_gripper.xml`
Evidence: `ur5e_setup_mujoco.png`, `assets/mujoco/`. ManiSkill3 trialled then dropped.

Sat 12 Jul — Fixed-camera view (today)

- [x] UR5e scene with a fixed third-person camera (matches the real eye-to-hand rig)
- [x] Offscreen `mujoco.Renderer` captures RGB frames (240x320)
- [~] RGB-D / depth output
Depth render touched (`test_depth.png`); depth-to-3D conversion itself is Week 6 work.
- [x] Save a few frames as PNG to verify

Sun 13 Jul — Demonstration recorder

- [x] DemoRecorder captures joints, TCP pose, gripper state, image at ~20 Hz
- [x] save_episode() writes per-field arrays to HDF5
- [x] Replay path: replay_episode.py reads HDF5 back to MP4
- [x] Test by scripting a pick motion and recording it (record_replay_episode.py)
- [x] Visualise the trajectory in matplotlib; verify images
- [x] Bonus beyond plan: teleop host/client over sockets, DLS IK solver, gripper-attach script

WEEK DELIVERABLES

- [x] OpenVLA architecture understood (can explain it to a friend)
- [x] MuJoCo working with a UR5e and a fixed camera (replaces 'ManiSkill3 working')
- [x] DemoRecorder working; at least one episode saved and replayed
- [] Open X-Embodiment paper read (only remaining Week-4 item)

WEEK 5 · 14–20 Jul 2026 — SmolVLA + LeRobot (Architecture B training)

Done 0 · In progress 0 · To do 12 · 12 items

Plan note — data capture (DemoRecorder) is already done; this week is the training half. Add one bridging piece your architecture doc lists as 'planned': an HDF5 to LeRobot/RLDS converter turning recorded episodes (absolute states, axis-angle) into training format (action deltas, Euler). Nothing consumes your episodes yet.

Mon 14 Jul — LeRobot installation

- [] Clone huggingface/lerobot; install smolvla extras; log into HF Hub; explore the layout

Tue 15 Jul — SmolVLA paper (first half)

- [] Sec 1–3: ~450M params (vs 7B OpenVLA), SmolVLM-2 backbone + Action Expert, async inference

Wed 16 Jul — SmolVLA paper (second half)

- [] Sec 4–6: training data, key ablations; write a 1-page summary

Thu 17 Jul — LeRobot SmolVLA tutorial

- [] Open the tutorial; download the dataset; print shape + episode count; visualise frames

Fri 18 Jul — Cloud GPU training setup

- [] Modal Labs or Colab Pro; training script on the cloud GPU; confirm dataset access; 1-step dry run

Sat 19 Jul — Run the fine-tune

- [] Launch the full run (1–3 h); monitor loss; save the checkpoint; note cost + time

Sun 20 Jul — Inference + RTC paper

- [] Load checkpoint; inference on a held-out episode; predicted vs ground-truth
- [] Read the training script line by line; read RTC (Black et al. 2025): async inference + chunking

WEEK DELIVERABLES

- [] Fine-tuned SmolVLA checkpoint saved (cloud GPU)
- [] SmolVLA paper understood + 1-page summary
- [] RTC paper read
- [] Inference working: predicted vs ground-truth comparison

WEEK 6 · 21–27 Jul 2026 — Perception + modular pipeline + LLM planner (Architecture A)

Done 0 · In progress 0 · To do 15 · 15 items

Plan note — prototype the whole modular slice in MuJoCo, not URSim. MuJoCo provides the RGB-D camera, depth, and gripper physics this stack needs; URSim has none of that. Build primitives on the shared controller interface so nothing needs rewriting for hardware. This is Architecture A in full, now central rather than peripheral.

Mon 21 Jul — Open-vocabulary detection

- [] Run OWL-ViT / GroundingDINO on a sample image; text query to boxes + confidence; threshold; visualise
The crux vs closed-set YOLO: it can name and localise novel objects.

Tue 22 Jul — Depth to 3D position

- [] Box-centre pixel + depth + camera intrinsics to camera-frame XYZ; small NumPy function
You already render depth in MuJoCo; feed it straight in.

Wed 23 Jul — Hand-eye calibration

- [] Fixed-camera transform; object-in-camera to object-in-base (one 4x4 multiply); NumPy
Reuses transforms.py (currently empty; populate it first).

Thu 24 Jul — Scripted motion primitives

- [] pick(xyz): above, descend, close, lift. place(xyz): above, descend, open, retract
- [] Build on the shared controller; test pick-and-place in MuJoCo with a hardcoded object position

Fri 25 Jul — Anthropic SDK + tool calling

- [] API key; hello-world Claude call; tool-use docs; define pick/place/move as tools; note limits + cost

Sat 26 Jul — LLM planner over primitives

- [] System prompt with skills + detected objects; LLM returns an ordered tool-call plan (structured)
- [] Parse and dispatch to primitives; test 'put the red block in the bin' -> detect -> plan -> execute

Sun 27 Jul — Closed-loop modular slice + write-up

- [] Wire the full slice: frame -> detect -> depth-to-3D -> positions -> LLM plan -> execute in MuJoCo
- [] Add a re-detect step before each grasp (closed loop); document failures; update README
Open-loop first; add the re-detect step once it runs end-to-end.

WEEK DELIVERABLES

- [] Open-vocabulary detector returning boxes on test images
- [] Depth-to-3D + hand-eye transform implemented (NumPy)
- [] Scripted pick/place primitives working in MuJoCo
- [] LLM planner sequencing primitives via tool calls
- [] Closed-loop modular slice running end-to-end in simulation

WEEK 6.5 · 28 Jul – 4 Aug 2026 — Refinement, consolidation, sign-off

Done 0 · In progress 1 · To do 10 · 11 items

Mon 28 – Tue 29 Jul — Refine the proposal

- [] Re-read the proposal end to end; check NTU EEE format; update the architecture diagram to MuJoCo; tighten methodology

Wed 30 Jul — Send for sign-off

- [] Send to supervisor; prepare 3–5 questions; confirm VLA choice, lab GPU access, budget

Thu 31 Jul — Consolidate the codebase

- [] Tidy repo + README; ensure URController, DemoRecorder, detector, planner each have a runnable demo

Fri 1 – Sat 2 Aug — Catch up + revise

- [] Absorb slippage from Weeks 1–6; revise on feedback; re-read the 5 core papers

Sun 3 – Mon 4 Aug — Final prep for FYP start

- [] 1-page current-state summary for FYP Week 1; priority list for the first two FYP weeks; set a review cadence

WEEK DELIVERABLES

- [] Proposal refined and sent for sign-off
- [] Codebase consolidated: URController, DemoRecorder, detector, planner all runnable
- [] Fine-tuned SmoIVLA checkpoint (tutorial dataset)
- [] Closed-loop modular slice working in simulation
- [~] Notes on at least 5 deeply-read papers (RT-1, RT-2, OpenVLA done; OXE, SmoIVLA, RTC remaining)
- [] 1-page current-state summary + Week 1 priority list